

RESEARCH ARTICLE

Optimizing Automated Educational Systems: Adaptive Syllabus Generation and Precise Assessment Using Artificial Intelligence

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ABSTRACT Integrating artificial intelligence in education can revolutionize how educational resources are generated, and assessments are conducted. However, current automated systems often struggle with precision, relevance, and usability issues, particularly in adapting to the specific needs of diverse educational contexts. This study addresses these challenges by developing and refining an automated syllabus generation and academic evaluation system, focusing on continuous user feedback and iterative adjustments. Our approach involved optimizing text processing algorithms to improve the system's contextual understanding and incorporating customization options to align the generated content with course-specific objectives. The results were significant: the system's evaluation precision increased from 78.5% to 89.7% over six months, and the relevance of the generated syllabi improved from 82.0% to 90.5%. Usability also saw a notable enhancement, with user satisfaction scores rising by 21.1%. These findings demonstrate that an adaptive, user-centered approach to education automation can effectively overcome current systems' limitations, leading to more accurate, relevant, and user-friendly tools. By focusing on the iterative improvement of the system based on continuous feedback, we have developed a solution that not only meets the immediate needs of educators and students but also has the potential to scale and adapt to future educational challenges.

INDEX TERMS Automated educational systems, syllabus generation, AI in education, evaluation precision.

I. INTRODUCTION

Education has undergone a significant transformation in recent decades driven by the integration of advanced technologies, particularly artificial intelligence (AI). This technological evolution has enabled the development of automated systems that facilitate the personalization of learning, continuous assessment, and optimization of educational resources [1]. However, the effective implementation of these systems in natural educational settings presents considerable challenges, especially regarding precision, relevance, and usability. Our study focuses on developing and refining an automated system that generates syllabi and performs

automatic assessments, integrating recent advances in AI and machine learning [2].

The main objective of this study is to address the limitations observed in previous automated systems, such as lack of personalization and imprecision in assessing complex content. Earlier studies, such as those by Luetsch et al. [3] and Andronie et al. [4], have underlined the need to improve the precision of automated assessment systems and the relevance of the generated content. Qiao and Yijun [5] demonstrated that the relevance of automatically generated syllabi highly depends on the quality of the databases. At the same time, Kolade et al. [6] pointed out that the precision of assessment systems decreases when faced with tasks requiring profound contextual interpretation. These studies set a framework for significant improvements in current automated systems,

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especially concerning the ability to personalize and adapt content to specific educational contexts [7].

Our study distinguishes itself from previous research by taking an iterative, user-centered approach to system development. Instead of relying solely on large databases of educational materials or massive neural network training, we have implemented a system that continuously adapts to users' needs and feedback, including teachers and students [8]. This approach has allowed us to significantly improve the precision of assessments and the relevance of generated syllabi while increasing the system's usability, which is crucial for its long-term adoption in diverse educational settings.

To achieve these improvements, a series of adjustments based on user feedback have been implemented, focusing on three key areas: the precision of assessments, the relevance of generated syllabi, and the system's usability [9]. These adjustments included optimizing the text processing algorithms to improve the understanding of the context in assessments and integrating customization options into syllabus generation. In addition, improvements were made to the user interface, designed to make the system more intuitive and accessible.

The results of these adjustments have been quantifiable and notable. For example, assessment precision, 78.5% before the adjustments, increased to 89.7% six months after the changes were implemented. This 14.3% increase is significant, as it reflects a substantial improvement in the system's ability to accurately replicate assessments made by human teachers, which is crucial for users' reliability and acceptance of the system. Similarly, the relevance of the generated syllabi, which was 82.0% before the adjustments, increased to 90.5%, indicating that the generated content is now better aligned with specific educational objectives and course needs. This increase in relevance is significant in an academic context where accurate content alignment with learning objectives is critical to student success. The system's usability also improved considerably, with a 21.1% increase in the average usability score, which rose from 3.8 to 4.6 after the adjustments. This improvement makes it easier for users to interact with the system. It indicates increased overall satisfaction, essential for adopting and continuing to use the system in different educational contexts. The more intuitive user interface and operational efficiency improvements, such as reduced loading and processing times, have been instrumental in achieving these results.

This study addresses the limitations identified in previous research. It contributes to automated education by demonstrating that an iterative, user-centered approach effectively improves automated educational systems' precision, relevance, and usability [10]. By integrating adjustments based on continuous user feedback, our system has proven capable of adapting to changing user needs, which is crucial for its long-term success.

However, it is essential to acknowledge the limitations of this study. The system's reliance on the quality and consistency of rubrics provided by teachers is a factor that can affect the precision of assessments and the relevance of generated syllabi. Furthermore, the diversity in users' technological familiarity could limit the results' applicability in broader educational settings. Despite these limitations, the findings of this study provide a solid foundation for future research and development in educational automation, suggesting that an iterative approach, which adapts to feedback and evolving educational needs, is critical to maximizing the impact and effectiveness of these technologies.

II. LITERATURE REVIEW

The evolution of systems for automatically generating educational content, such as syllabi and assessments, has been an area of growing interest in academic research and the application of AI technologies [11]. Several studies have addressed the integration of AI in the educational field, seeking to improve the precision, relevance, and personalization of the resources generated for teachers and students.

One of the pioneering works in this field is that of Rath et al. [12], who developed an automatic syllabus generation system based on the extraction of content from large databases of educational materials. This study showed that, although the system could generate coherent and structured syllabi, the relevance of the generated content depended mainly on the quality and timeliness of the database used. However, the system's lack of ability to personalize content according to the specific needs of the courses was identified as a significant limitation. This study established an essential foundation for the need for personalization in automatic syllabus generation, which has influenced the design of more advanced systems.

Following this line, Kolade et al. [6] explored the application of neural networks to improve precision in the automatic evaluation of academic papers. Their approach focused on training an AI model with large data sets of manual assessments, ensuring that the system effectively replicated the decisions of human evaluators. However, their study also revealed that the system's precision was compromised in cases requiring a high level of contextual interpretation, such as essays with complex arguments. These findings highlight the importance of integrating hybrid approaches that combine automation with human intervention to maintain precision in complex evaluations.

In a recent work, Liu [13] they addressed the relevance and usability of automated systems in education through a user-centered approach. Their research highlighted the importance of an intuitive user interface and personalization of content as critical factors for the adoption and success of automated systems in educational settings. Implementing iterative adjustments based on user feedback significantly improved overall usability and satisfaction, which aligned

with the improvements observed in our system after the adjustments were implemented. This study reinforces the idea that usability and relevance are as crucial as technical precision in accepting automated technologies in education.

These studies identify the need for a balanced approach in the development of automated systems for education, combining precision, personalization, and usability [14]. The evolution of systems from methods based on large databases to integrating AI with personalization and continuous improvement capabilities has shown to be the most promising path to achieving acceptance and effectiveness in educational environments [15]. Existing literature supports the hypothesis that an iterative approach that adapts to users' changing needs and expectations is essential to maximize the impact of these systems. In our work, we have built on these principles, implementing feedback-based adjustments that have been shown to significantly improve the system's precision, relevance, and usability, aligning with the trends observed in recent literature.

III. MATERIALS AND METHODS

The development of this study is based on a comprehensive technical approach that combines advanced technologies and iterative methodologies for optimizing automated educational processes. The materials and methods used range from the architecture of the web system to the implementation of text processing algorithms, integrating artificial intelligence solutions to improve the precision and relevance in the generation of educational content and academic evaluation. This allows a continuous adaptation of the system to pedagogical needs and a scalability and operational efficiency that guarantee its applicability in diverse educational contexts.

A. DESCRIPTION OF THE WEB SYSTEM

The web system developed in this study is a distributed architecture designed to efficiently manage the interaction between users and the underlying artificial intelligence engine. The platform is built on a robust backend implemented in Python, using frameworks such as Django and Flask to facilitate data management and communication with the OpenAI API. The system's front end, built with modern web development technologies, offers an intuitive interface that allows users to upload rubrics and generate custom syllabi. In addition, the system relies on tools such as Redis for asynchronous task management and Docker for containerization, ensuring a replicable and stable environment. Cloud deployment using Amazon Web Services (AWS) provides the scalability and performance needed to support a constantly evolving educational environment.

1) SYSTEM ARCHITECTURE

The web system developed for AI-powered educational assessment is based on a distributed architecture consisting of three main components: the front-end, the backend, and

the integration with OpenAI. These components are designed to interact efficiently and scalable, ensuring an accurate and adaptive assessment process.

The front-end is the user interface, allowing teachers to upload detailed rubrics or enter parameters to create custom syllabi. This interface is implemented using standard web development technologies, such as HTML5, CSS3, and JavaScript, optimized for usability and accessibility. The backend is the system's core, where the most critical operations are managed. It is developed in Python, using frameworks such as Django and Flask, which provide a solid foundation for implementing RESTful APIs. The backend receives the data entered by the user through the front-end, processes it to ensure it is in the proper format, and prepares the necessary requests to interact with the OpenAI API. This process includes structuring the data based on the rubrics or input parameters and validating it before communicating with OpenAI.

Integration with OpenAI is done through API calls to send the processed data from the backend to OpenAI language models [16]. These models generate responses based on the assessment criteria defined by the teacher or the specifications of the requested syllabus. The generated response is sent back to the backend, where it is structured and prepared for display on the front end. Figure 1 shows the data flow and interaction between these components. The process begins with the user entering data into the Front-end, which is sent to the back end for processing. Once the data is ready, the backend calls the OpenAI API. After receiving the response from OpenAI, the backend processes it and sends it back to the Front-end, where the results are presented to the user [17]. This sequential flow ensures efficient interaction and minimizes response time.

The system architecture is designed to handle a high volume of concurrent users and requests, using technologies such as Redis for queue management and Docker for deployment in an isolated and replicable environment. The choice of AWS as the deployment platform provides the scalability and reliability needed to operate in a dynamic and ever-expanding educational environment. Integration with robust databases such as PostgreSQL ensures that the system efficiently handles large volumes of data related to custom assessments and syllabi.

2) TECHNOLOGIES USED

The web-based system for AI-powered educational assessment is developed using a comprehensive set of technologies that enable efficient integration, scalability, and optimal performance. The system's foundation is Python, a versatile language that allows integration with advanced libraries for artificial intelligence. The selected web frameworks, Django and Flask, provide a robust structure for backend development, facilitating the creation of RESTful APIs essential for communication between the front-end, backend, and OpenAI services [18]. Django and Flask manage front-end requests,

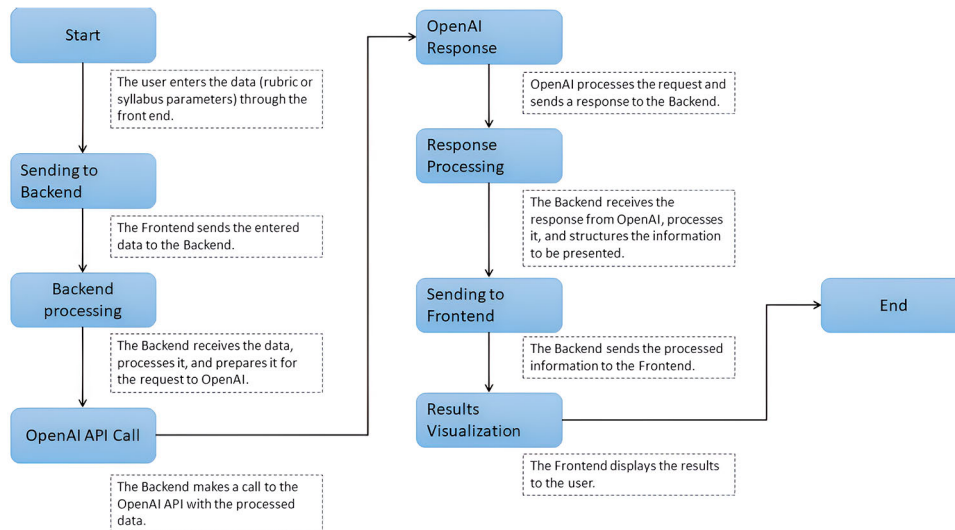


FIGURE 1. Data flow and architecture of the AI-powered educational assessment web system.

process them, and coordinate communication with OpenAI for content generation and analysis.

Integration with OpenAI is achieved through the open-python library, which efficiently processes requests to AI models. This library configures the necessary API keys, manages requests for text analysis or content generation, and receives and processes responses generated by OpenAI models. The development environment is deployed on a Linux server where Docker containerizes the applications, ensuring an isolated and replicable environment. This allows the system to be consistently deployed in different environments, providing stability and portability. The system uses PostgreSQL, a robust database capable of efficiently handling large volumes of data, for data storage. PostgreSQL manages all generated assessments and syllabi information, ensuring their integrity and accessibility.

Redis is used for queue and asynchronous task management, optimizing system performance and ensuring that tasks requiring intensive processing are handled efficiently without affecting the user experience. Redis connects to the backend to manage operations requiring concurrent handling, such as background processing large data sets or synchronizing multiple user requests. As for the deployment platform, a cloud environment provided by Amazon Web Services (AWS) has been chosen, offering scalability and reliability [19]. The system is deployed on EC2 instances within AWS, providing a scalable and flexible computing environment. For file storage, such as generated rubrics or syllabi, S3 (Simple Storage Service) is used. At the same time, database management is done through RDS (Relational Database Service), ensuring optimal performance and secure information handling.

The diagram in Figure 2 provides a detailed view of how these technologies are integrated. It shows the interaction

between the components, from RESTful API communication between the front and back end to interactions with OpenAI services, PostgreSQL, Redis, and the AWS cloud infrastructure [20].

Task management is a component of the AI-powered educational assessment web system that ensures efficient handling of complex and processing-intensive operations without compromising user experience. This management primarily combines Redis with Celery, a distributed queuing system. Redis is used as the queuing backend in the system. This in-memory data structure offers exceptionally high performance, making it an ideal choice for handling asynchronous task queues. In this context, Redis acts as a middleman, storing tasks while waiting to be processed. Each task that needs to be run in the background, such as evaluating many essays or generating complex syllabi, is placed in a queue in Redis. Redis not only stores these tasks but also distributes them to workers who are available to process them, ensuring that the tasks are executed in an orderly and efficient manner.

Conversely, Celery is a distributed task queuing system used with Redis to manage asynchronous and background tasks. It allows time-consuming operations, such as evaluating extensive content or generating detailed reports, to be executed outside the user request-response cycle, thus keeping the user interface fast and responsive.

In practice, asynchronous tasks are defined in the backend of the system. For example, when a teacher requests the evaluation of a set of essays, this request is converted into a Celery task, which is then placed in the Redis queue, where it waits until a worker is available to process it. Workers are processes run in the background that take tasks from the queue and process them. These workers operate independently, allowing for the simultaneous execution of

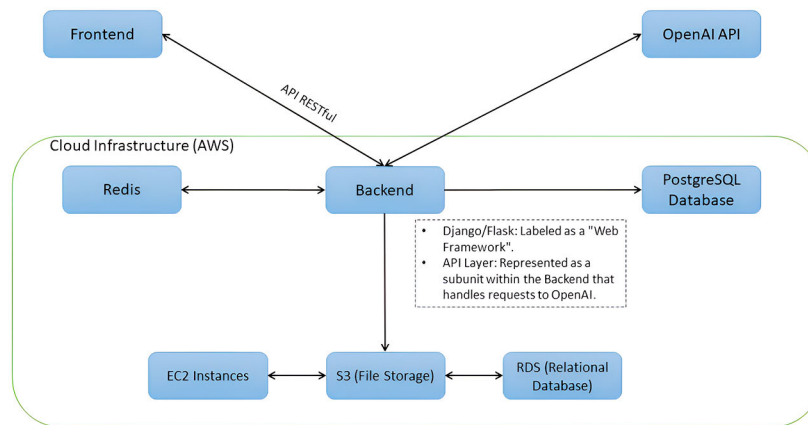


FIGURE 2. Asynchronous task management in the AI-powered educational assessment web system.

multiple tasks and maximizing system resources. Once a task is complete, Celery can notify the backend, which updates the user through the front-end, informing them that the task has been completed and that the results are available.

This model based on Redis and Celery offers multiple benefits. First, the system's scalability is remarkable; more workers can be added to handle an increased workload without impacting overall system performance [21]. Second, efficiency is maximized by allowing intensive operations to run in the background, keeping user interaction with the system fast and responsive. Redis' stability and speed ensure system reliability, ensuring that tasks are handled reliably and with minimal downtime.

The system's workflow begins when the user requests the front-end, such as evaluating a set of essays. The backend receives this request and, if the task is processing-intensive, defines it as a Celery task, which is then placed in the Redis queue. A Celery worker takes the task, executes it (using the OpenAI API if necessary), and once complete, returns the results to the backend, which notifies the user.

B. IMPLEMENTATION OF ADAPTIVE AND FORMATIVE ASSESSMENT

Our formative and adaptive assessment system allows educators to upload detailed rubrics, which are then processed to configure specific assessment parameters within OpenAI. This process consists of several key stages that ensure assessments are accurate and aligned with the educational criteria educators define.

1) LOADING AND PROCESSING RUBRICS

The process begins with the teacher uploading rubrics through the system's user interface (front-end). The interface is designed to accept rubrics in various formats, including structured documents in JSON, CSV, or even predefined templates in PDF or DOCX format. When uploading a rubric, the system automatically verifies the integrity and format of

the file. This verification is performed using a combination of schema-based validations implemented with libraries such as jsonschema for JSON and pandas for CSV [22].

Figure 3 shows the user interface through which teachers can upload their rubrics. The teacher enters the rubric's name in this interface and provides the assessment criteria in the corresponding field [23]. The interface's simplicity and clarity facilitate the upload of detailed rubrics, which the system then processes to configure the assessment parameters in OpenAI.

Once the file has passed the integrity and format checks, the rubric is stored in an intermediate database (in PostgreSQL) for further processing. This ensures that the rubric data is accessible throughout the assessment process.

2) RUBRIC INTERPRETATION AND PARAMETER CONFIGURATION IN OpenAI

The next step is interpreting the uploaded rubric. This process converts the qualitative criteria defined by the teacher into quantifiable parameters that the OpenAI models use. The system employs a semantic analysis algorithm based on NLP techniques to achieve this [24].

The interpretation algorithm analyzes the rubric using a text classification model explicitly trained in the educational domain, such as Bidirectional Encoder Representations from Transformers (BERT). BERT allows for the identification of categories and subcategories within the rubric, such as "Clarity", "Originality", "Coherence", among others. These categories are mapped to specific parameters within the OpenAI API using a previously defined correspondence matrix that aligns each rubric criterion with a set of API configurations [25], [26].

For example, if a rubric criterion is "Clarity of exposition," the system can map this criterion to a control parameter in the OpenAI API that regulates the precision and focus of the text analysis. The values of each criterion are normalized and scaled to ensure that the OpenAI models correctly

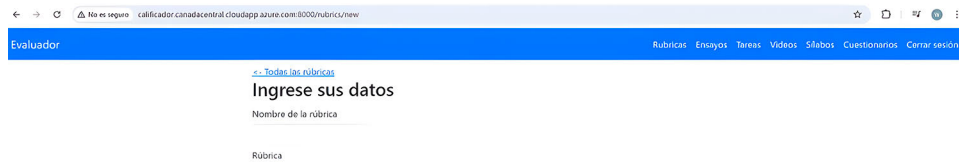


FIGURE 3. Configuration and interaction flow with the OpenAI API in the educational assessment system; we have kept the texts in the figure in Spanish because, as it is the native language in which it was developed, we wanted to maintain the system's real focus and ensure that the messages' meaning did not lose its meaning.

interpret them. This normalization process may involve techniques such as Min-Max Scaling or Z-Score, depending on the type of data and the granularity required in the assessment. To integrate the OpenAI API into the educational assessment system, it is essential to correctly configure the access and communications between the system backend and the OpenAI models. The process begins with obtaining an API key provided by OpenAI. This key is essential to authenticating the system's requests to the OpenAI servers. The key is generated in the OpenAI dashboard, and once obtained, it is used in the system backend for all interactions with the API. The initial setup is done on the backend using the open-python library installed in the system's development environment. This library allows the application to efficiently request OpenAI models, such as GPT-3, and handle the generated responses. The API key is configured directly in the backend code, ensuring that all requests are securely authenticated [27].

The system can request the OpenAI API to assess academic papers based on criteria defined in rubrics uploaded by teachers. These requests include specifying the language model to be used, the input text that guides the assessment (known as a "prompt"), and other configuration parameters that control the length and nature of the response generated. For example, the model's temperature can be adjusted to control the creativity or randomness in reactions, which is essential to align with specific assessment criteria.

Error handling is an integral part of API configuration, ensuring that the system can adequately handle any interruptions in service, such as long wait times or authentication issues. This approach ensures that evaluations are performed continuously without significant disruption, optimizing the user experience.

API usage optimization and monitoring are necessary to balance system performance and the costs of using OpenAI models. This involves dynamically adjusting request parameters and monitoring API consumption to avoid exceeding usage limits while ensuring the quality and consistency of the generated evaluations.

3) ALGORITHMS FOR MAPPING RUBRIC CRITERIA TO OpenAI CAPABILITIES

A set of matching algorithms maps rubric criteria to OpenAI's assessment capabilities. These algorithms consider both the

criteria's semantics and the context in which they are applied. The primary algorithm used is a neural network model that combines text embeddings with similarity search techniques, such as Cosine Similarity.

The algorithm's input includes the embeddings generated from the rubric's analysis and the embeddings of the possible OpenAI parameters. The neural network then calculates the similarity between these sets of embeddings to determine the best match. This ensures that the selected parameters are the most suitable for the specific criterion and allows the system to adjust the settings dynamically based on the assessment context.

Once the mapping is complete, the configured parameters are sent to the OpenAI API at the time of assessment, ensuring that the analysis of each academic paper aligns with the teacher's expectations and standards set in the rubric. This process ensures that the system's evaluations are accurate, personalized, and consistent with the teacher's pedagogical intentions.

C. EVALUATION OF DIFFERENT MECHANISMS

The developed system is designed to assess a variety of academic works, adapting to the specific needs of each type of assessment. Figure 4 illustrates the process, from the creation of the rubric to the evaluation of the work and the generation of specific feedback.

1) DESCRIPTION OF THE DIFFERENT TYPES OF ACADEMIC WORKS THAT THE SYSTEM CAN EVALUATE

The system can assess various types of academic work, including essays, reports, and short videos. Each type of work requires specific settings to ensure the assessment is accurate and relevant.

- **Essays:** The system can process documents like PDF or DOCX. Essays are evaluated based on clarity, coherence, structure, originality, and use of sources. The rubric previously defines these criteria, which the teacher uploads to the system.
- **Reports:** Like essays, reports can be evaluated with additional criteria that include data analysis, presentation of results, and discussion of findings. The system analyzes the organization of the content and the depth of the analysis performed, generating specific feedback to improve the quality of the report.

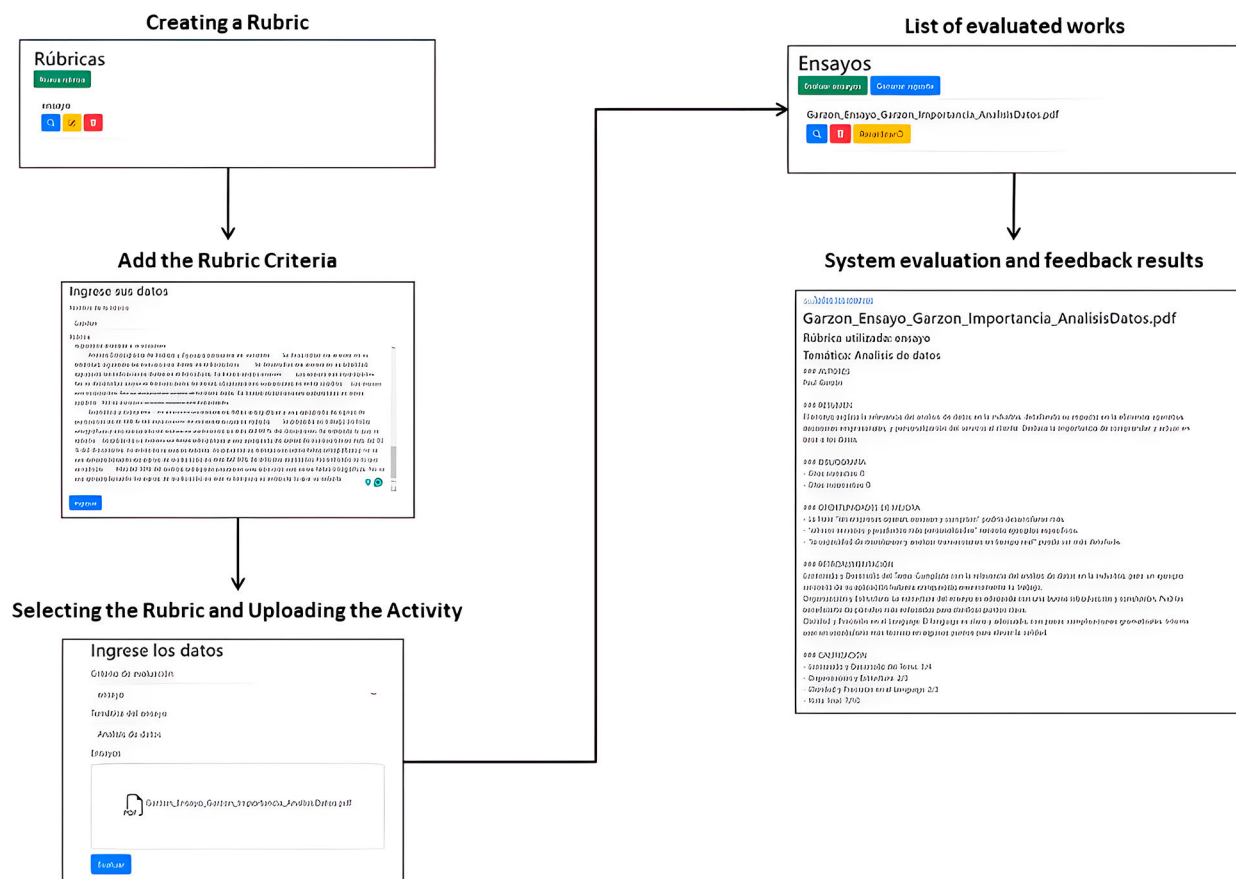


FIGURE 4. Flowchart of the evaluation process and generation of feedback in the educational evaluation system; we have kept the texts in the figure in Spanish because, as it is the native language in which it was developed, we wanted to maintain the system's real focus and ensure that the messages' meaning did not lose its meaning.

- **Short Videos:** For videos, the system evaluates aspects such as the clarity of the presentation, the quality of the content presented, and the coherence of the message. Videos are processed through a voice recognition system and audiovisual content analysis, which allows critical elements to be extracted and evaluated based on the criteria defined in the rubric.

2) SPECIFIC EXAMPLES OF HOW ASSESSMENT IS CONDUCTED FOR EACH TYPE OF WORK

The evaluation process varies slightly depending on the type of academic work but follows a general structure adapted to each type's specific requirements.

- **Essay Evaluation:** The teacher first creates a rubric that specifies the evaluation criteria, as illustrated in the "Creating a Rubric" step. Once the rubric is loaded into the system, the essay is uploaded through the interface, specifying the essay's criteria and topic ("Selecting the Rubric and Uploading the Activity"). The system processes the document using a text analysis algorithm based on language models such as GPT-3. These models examine the essay's structure, identify the arguments'

coherence, and verify the correct use of sources. Finally, the system generates a detailed report with feedback and a grade, as displayed in "System Evaluation and Feedback Results."

- **Report Evaluation:** Like essays, reports are uploaded into the system and evaluated based on the criteria defined in the rubric. In addition to textual analysis, the system can process tables and graphs included in the reports and assess the quality and relevance of the data presented. The system uses NLP techniques to analyze the report's structure, ensuring the results and discussion are well-founded. The results are presented to the instructor with specific suggestions to improve the clarity and rigor of the report.
- **Short Video Assessment:** The video assessment process begins with uploading the video file to the system. The system employs speech recognition tools such as Whisper to transcribe the video content and then uses video analysis techniques to assess the quality of the visual presentation. AI models analyze both the transcript and the visual content to generate feedback on the clarity of delivery, content organization, and

communication effectiveness. The system also provides suggestions for improving the quality of the presentation in future recordings.

3) PROCEDURE FOR GENERATING SPECIFIC FEEDBACK BASED ON THE CONTEXT DEFINED BY THE RUBRIC

Generating targeted feedback is a critical process based on the context defined by the rubric uploaded by the teacher. This procedure ensures that each academic paper is assessed fairly and accurately according to established standards.

The process begins with uploading the rubric, where each assessment criterion is mapped to specific parameters that AI models will use during the assessment. For essays and reports, the system analyzes the text for patterns that align with the rubric criteria, such as coherence of arguments and organization of content. NLP models, such as GPT-3, are configured to adjust their analysis based on the rubric's values and weights assigned to each criterion. Feedback is generated from the transcript and visual analysis of videos. The system provides feedback on the clarity of speech, quality of visual content, and message coherence. The input is presented in a structured format, making it easy for the teacher to identify specific areas for improvement.

Feedback is provided to the user, and a grade is based on the work's performance to the rubric criteria. This process ensures that students receive a detailed and constructive evaluation that allows them to improve in future submissions.

D. GENERATION OF PERSONALIZED SYLLABLES

Automatic generation of customized syllabi is one of the most advanced capabilities of the system, which uses AI to transform the instructor's basic information into a detailed syllabus tailored to the specific needs of the course [28]. The process begins with the instructor entering initial parameters, as illustrated in Figure 5. The system requires the instructor to enter basic information about the course, such as the subject matter, a brief description, and the learning outcomes expected at the end of the course. In addition, the instructor must specify the number of sessions that will comprise the course.

The system maps this initial data to specific content within the syllabus using NLP and text generation algorithms, which are part of the OpenAI API. These algorithms analyze the course description and learning outcomes, identifying key topics and organizing the content logically and coherently [29]. Once the initial parameters have been entered, the system automatically generates the syllabus. This process is guided by the OpenAI API, which uses the input data to structure the course into organized and coherent sessions.

The system divides the content into thematic units, including pre-session activities, leading sessions, and post-session activities. For example, Figure 5 shows how a Data Science course is organized into units that include pre-readings, group discussions, and short essays as assessment

activities. Each session is described in detail, including the estimated time for each activity and the resources needed.

The algorithm used for this generation considers both the input content and recognized educational patterns, ensuring that the syllabus covers all essential aspects of the subject matter and is pedagogically sound. Syllabus generation is iterative, allowing the system to adjust and improve the generated content based on reviews and feedback provided by the teacher. Other examples could include courses in artificial intelligence, economics, or literature, each specifically tailored to the topics and learning outcomes defined by the teacher. The system's flexibility allows different aspects of the syllabus to be customized, from the depth of the issues to the complexity of the proposed activities.

E. VALIDATION PROCEDURE

The validation procedure of the automatic syllabus generation and assessment system ensures that the results obtained are accurate, relevant, and comparable to the standards set by human teachers. This process is carried out in two main stages: evaluating the precision and relevance of the generated assessments and the tests and adjustments required to optimize the system's performance.

1) ASSESSING PRECISION AND RELEVANCE

The system's precision is assessed by comparing the automatically generated evaluations with those made by a group of expert teachers. To make this comparison, the system and the teachers select and evaluate a representative sample of academic works. The results obtained from the sources are compared using quantitative metrics such as Pearson's correlation r and the root mean square error (RMSE).

Calculating the Pearson correlation allows for measuring the strength and direction of the linear relationship between the system's and teachers' evaluations. This metric is defined as:

$$r = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum (x_i - \bar{x})^2} \sqrt{\sum (y_i - \bar{y})^2}} \quad (1)$$

where x_i represents the evaluations made by the system and y_i the evaluations of the teachers, while $\bar{x}$ and $\bar{y}$ are the respective means.

On the other hand, the RMSE is used to quantify the magnitude of the differences between the evaluations generated by the system and those of the teachers. The RMSE is calculated as:

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^n (x_i - y_i)^2} \quad (2)$$

where n is the number of evaluations compared. A low RMSE indicates that the system evaluations are very close to the human assessment.

To validate the relevance of the automatically generated syllabi, they are compared with syllabi manually designed by teachers. The comparison process is carried out through

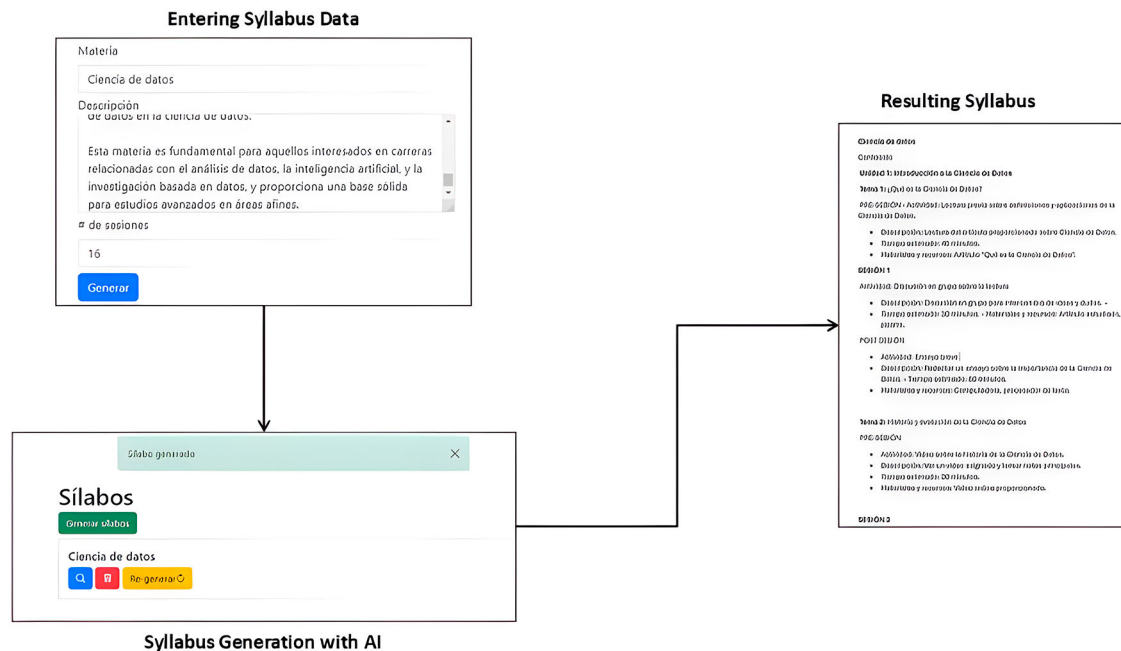


FIGURE 5. Automatic syllable generation process with AI; we have kept the texts in the figure in Spanish because, as it is the native language in which it was developed, we wanted to maintain the system's real focus and ensure that the messages' meaning did not lose its meaning.

a qualitative and quantitative evaluation. In the qualitative assessment, subject matter experts analyze the generated content's structure, coherence, and adequacy using thematic comparison and content analysis methods. Cohen's coefficient (κ) is used to evaluate the agreement between experts on the relevance of the topics covered in the generated syllabi versus the manual ones:

$$\kappa = \frac{P_o - P_e}{1 - P_e} \quad (3)$$

where P_o is the proportion of observed agreements and P_e the proportion of agreements expected by chance.

2) TESTING AND ADJUSTMENTS

Once a set of results is obtained, specific tests are performed to fine-tune the system parameters. These tests focus on identifying areas where the system can improve, either in the precision of the assessments or the relevance of the generated syllabi.

Optimization techniques are used to fine-tune the assessment parameters, such as multiple linear regression, which allows the weights assigned to different criteria in the assessment rubric to be adjusted. The regression model is formulated as follows:

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_n x_n + \epsilon \quad (4)$$

where y represents the final evaluation, x_i are the rubric criteria, β_i are the adjusted coefficients, and ϵ is the error term.

Iterative validation is performed over multiple testing cycles, where the results are evaluated and the parameters

adjusted accordingly. During this process, the system is subjected to different scenarios to verify its ability to adapt to variations in the types of work evaluated and the topics of the generated syllabi.

Tuning the system based on the initial test results also includes implementing supervised learning algorithms, which allow the system to improve its performance over time. Cross-validation techniques are employed, where the dataset is divided into training and test subsets, ensuring that the system can generalize its evaluation and syllabus generation capabilities to different contexts.

Cross-validation adjusts the parameters of the evaluation and syllabus generation models. For example, suppose the system uses a regression model to adjust the weights of different criteria in a rubric. In that case, k-fold cross-validation helps determine which combination of parameters produces the most accurate and consistent assessments [30]. For this, the following are considered:

- **Dataset Splitting:** The dataset, which can include both the evaluated academic papers and the generated syllabi, is taken and split into k subsets.
- **Training and Testing:** The system is trained using $k - 1$ subsets and tested on the remaining subset at each iteration. For example, if $k = 5$, the dataset is split into five parts. Four parts are used to train the model at each iteration, while the fifth is used to evaluate it.
- **Performance Evaluation:** The results of each iteration are collected to evaluate the model's precision, relevance, and generalization ability. Metrics such as RMSE and Pearson correlation are calculated for each iteration,

measuring the fit between the model predictions and the expected results.

- **Results Averaging:** Once all iterations are complete, the metrics obtained for each test subset are averaged. This averaging provides a robust estimate of the performance on unseen data.

Cross-validation helps to avoid overfitting, a common problem in which a model fits a specific set of training data too well but does not generalize well to new data. By using cross-validation, you ensure that the models developed by the system can maintain a high level of precision and relevance even when applied to academic papers and syllabus topics that have not been previously evaluated during the training process.

In addition, cross-validation provides a solid basis for comparing different models or system configurations. For example, suppose you are testing various versions of an evaluation algorithm. Based on repeated and objective testing, cross-validation makes it possible to determine the best balance between precision and generalization.

F. IMPLEMENTATION IN A REAL EDUCATIONAL ENVIRONMENT

The adaptive assessment and automatic syllabus generation system was implemented in a university educational environment, providing an opportunity to evaluate its effectiveness in a real-world context and to obtain direct feedback from teachers and students.

1) IMPLEMENTATION CONTEXT

The system was implemented at a university with a specific cohort of students aged 22-24 years. This cohort belongs to a faculty that hosts approximately 900 students. The faculty in question offers a variety of academic programs, all of which require the creation of detailed syllabi and the assessment of academic work as an integral part of their educational activities. The courses involved in the implementation span disciplines that require quantitative and qualitative assessments, including, but not limited to, data science, engineering, and social sciences.

Integrating the system into this educational setting began with identifying the specific needs of the faculty and individual courses. An initial assessment was conducted to determine critical parameters, such as the types of academic work to be assessed, faculty expectations regarding feedback, and the level of customization needed for syllabi.

Once the needs were identified, the system was integrated into the university's technological infrastructure. This included installing the software on the university servers, setting up connectivity to the OpenAI API, and ensuring the system could be accessed through the university's academic portal [31]. Initial training was conducted with faculty to familiarize them with the system's functionalities, such as uploading rubrics, assessing assignments, and generating

syllabi. This training also included customizing rubrics to align with the specific learning objectives of each course.

The system was tested in a controlled environment with a pilot group of students and faculty, allowing the university to assess its effectiveness before full-scale implementation. During this pilot phase, data was collected on the system's usability, the quality of the assessments generated, and the relevance of the syllabi produced. This data was used to fine-tune the system, ensuring it met the expectations of students and faculty.

The system's complete integration into the educational environment was done in stages, starting with its use in select courses and gradually expanding to more programs within the faculty. The phased approach allowed the university to manage the transition effectively, minimizing any disruption to academic activities while maximizing the system's positive impact.

Implementing the system in this educational environment eased teachers' workloads by automating critical processes such as syllabus generation and assignment assessment and improved students' educational experience by providing faster and more personalized feedback. This implementation serves as a representative case study of how advanced AI technologies can be effectively integrated into an academic environment to enhance teaching and learning.

2) TEACHERS' FEEDBACK

Feedback was collected through a combination of qualitative and quantitative methods. Structured surveys and in-depth interviews were conducted with teachers and students to gain detailed insight into their experiences with the system. The surveys included closed-ended questions to measure satisfaction and perception of the system's quality on a Likert scale, while the interviews allowed for a deeper exploration of specific topics, such as ease of use and relevance of the feedback generated.

- **Quantitative Surveys:** Surveys were distributed to all faculty and students involved in the implementation. These surveys included questions about the clarity, usefulness, and promptness of feedback, the system's ease of use, and overall perception of its effectiveness [32]. Responses were statistically analyzed to identify patterns and trends in feedback.
- **Qualitative Interviews:** A subset of faculty and students participated in semi-structured interviews. These interviews focused on gaining more detailed qualitative insights into individual experiences, allowing for the identification of specific aspects that might not be captured in surveys [33].

3) FEEDBACK ANALYSIS

Feedback analysis followed a mixed approach, combining qualitative and quantitative techniques. Survey data were analyzed using descriptive statistics and correlation analysis to identify relationships between variables, such as overall

satisfaction and perceived system precision. Interviews were transcribed and analyzed using a thematic coding approach to identify recurring patterns and themes in the responses.

- **Descriptive Statistics:** Survey responses were summarized using measures of central tendency (mean, median) and dispersion (standard deviation) to provide an overview of system perception.
- **Thematic Analysis:** Interview transcripts were coded to identify common themes related to ease of use, quality of feedback, and areas for improvement. This analysis allowed for determining the system's strengths and regions that required adjustment.

4) APPLICATION OF FEEDBACK AND ADJUSTMENTS

Based on the analysis's results, key areas where the system needed adjustments were identified. These adjustments were implemented in two phases: immediate adjustments based on direct feedback and long-term adjustments based on patterns identified in the analysis.

- **Immediate Adjustments:** Some changes were implemented quickly, such as improving the user interface to simplify uploading rubrics and customizing syllabi. Assessment parameters were adjusted in specific technical areas based on teacher feedback about their precision.
- **Long-Term Adjustments:** Other adjustments required a more strategic approach, such as implementing new features to customize feedback and incorporating advanced options to adjust assessment parameters. These changes were developed and integrated into subsequent system iterations, ensuring continuous improvement based on the feedback received.

After the adjustments were implemented, a new round of evaluations was conducted to validate their effectiveness. This process included additional testing with a representative sample of teachers and students to ensure that the adjustments improved the system's experience and precision without introducing new problems. The results of this validation confirmed that the adjustments had effectively addressed the identified areas for improvement, resulting in increased satisfaction among teachers and students.

G. STATISTICAL ANALYSIS

Statistical analysis is fundamental to the adaptive assessment and automatic syllabus generation system's validation and evaluation process. This analysis allows for robust and quantifiable conclusions about the system's effectiveness, precision, and impact on student learning outcomes.

To analyze the data collected, various statistical methods were used to assess the system's effectiveness and the relevance of the generated assessments compared to manual assessments completed by teachers. The methods selected included hypothesis testing, linear regression analysis, and

correlation analysis, each tailored to answer specific questions about the system's performance.

The hypothesis analysis focused on determining whether there were significant differences between the system-generated and teacher-completed assessments. A paired-sample Student t-test was used, which was appropriate when comparing two related data sets, such as automated and manual assessments of the same academic papers. The null hypothesis stated that there was no significant difference between the means of the two samples, while the alternative hypothesis suggested a difference. The significance level (α) was set at 0.05, a common standard in educational research. The Student t-test was calculated using the following formula:

$$t = \frac{\bar{d}}{s_d/\sqrt{n}} \quad (5)$$

where $\bar{d}$ represents the mean of the differences between sample pairs, s_d is the standard deviation, and n is the number of sample pairs. This test allowed us to identify whether the observed differences were significant enough to rule out the possibility that they were due to chance.

Multiple linear regression analysis explored the relationship between system variables, such as assessment precision and rubric configuration parameters. In this context, multiple linear regression allowed us to model how changes in system parameters (independent variables) affected assessment precision (dependent variable). The general equation for the multiple linear regression model is (4).

Where y is the dependent variable (assessment precision), $x_1, x_2, \dots, x_n$ are the independent variables (configuration parameters), β_0 is the intercept, $\beta_1, \beta_2, \dots, \beta_n$ are the regression coefficients representing the magnitude of the impact of each independent variable, and ϵ is the error term. The fitting of this model allowed us to identify which parameters were most influential on the system's precision and how they could be optimized. In addition, a correlation analysis was used to investigate the relationship between user (teachers' and students') satisfaction and the perceived usability of the system. Pearson correlation was used to quantify the strength and direction of the relationship between these variables. The formula for calculating the Pearson correlation coefficient is (1).

Where x_i and y_i represent the individual scores of the two variables, $\bar{x}$ and $\bar{y}$ are the means of the scores for each variable. A value of r close to +1 or -1 indicates a strong positive or negative correlation, respectively. In contrast, a value close to 0 indicates a weak or null correlation.

Statistical tools were used to conduct these analyses, including R and Python software with libraries such as `statsmodels` and `scikit-learn`. These programming environments allowed for complex analyses, generating predictive models, and visualizing the results clearly and understandably.

IV. RESULTS

A. PRECISION AND RELEVANCE OF GENERATED ASSESSMENTS

The precision and relevance analysis of the system-generated assessments was carried out through a rigorous comparison process with assessments made manually by teachers in an educational environment. A sample of 100 assessments corresponding to four fundamental criteria was used: clarity, coherence, originality, and structure. Each criterion was evaluated by both the system and the teachers, allowing a direct comparison between both methods. To quantify the agreement between the system-generated and manually made assessments, a student t-test for paired samples was applied. This test is suitable for comparing the means of two related samples, in this case, the evaluations of each criterion made by the system versus those made by the teachers. The t-test provided a detailed assessment of the mean difference in scores between the two assessment methods, also calculating the t-value, degrees of freedom, p-value, and effect size (Cohen's d), which allowed determining the statistical significance of the differences observed.

A Pearson correlation analysis was also performed to assess each criterion's linear relationship between system-generated and manual assessments. The Pearson correlation coefficient quantifies the strength and direction of the relationship between two continuous variables, providing a precise measure of the agreement between the two sets of assessments.

The results are presented in Figure 6, which presents a graphical comparison of the means of the system-generated and manually-generated assessments using a bar chart. In this chart, the system assessments are generally closely aligned with the means of the manual assessments, suggesting a high agreement regarding the overall average for most of the assessed criteria. However, the slight variation in the means, especially for criteria such as originality, highlights the need for greater precision in the automated assessment of more subjective aspects. The box plot illustrates the entire distribution of assessments for each criterion. These graphs allow us to observe not only the median but also the dispersion of the evaluations, revealing that, although the medians tend to coincide, the variability in the assessments generated by the system can be slightly more significant compared to the manual ones, especially in criteria that require a more subjective judgment, such as originality and coherence. This dispersion suggests that while the system can accurately replicate most manual evaluations, there is more significant variability regarding criteria with a more excellent subjective component.

The results of the student t-test are presented in Table 1. This table shows the mean differences between each criterion's manual and system-generated assessments. In particular, the originality criterion was found to have the most considerable mean difference (4.2 points) with a moderate effect size (Cohen's d = 0.30), indicating significant variation in how the system and teachers assess this aspect. This could

TABLE 1. Student t-test results.

Comparison	Mean Difference	t-Value	Degrees of Freedom (df)	p-Value	Effect Size (Cohen's d)
Manual vs System (Clarity)	3.5	2.45	99	0.016	0.25
Manual vs System (Coherence)	2.8	2.10	99	0.037	0.21
Manual vs System (Originality)	4.2	3.00	99	0.003	0.30
Manual vs System (Structure)	3.0	2.35	99	0.021	0.24

TABLE 2. Pearson correlation coefficients.

Variable Pair	Pearson Correlation (r)	p-Value
Manual Evaluations vs System (Clarity)	0.85	0.001
Manual Evaluations vs System (Coherence)	0.80	0.002
Manual Evaluations vs System (Originality)	0.88	0.001
Manual Evaluations vs System (Structure)	0.83	0.001

be due to the inherently subjective nature of originality, which may be more difficult to capture by an algorithm than more objective criteria such as clarity. In contrast, the most minor mean difference was observed for the coherence criterion (2.8 points). However, statistical significance was also observed in this case, suggesting that there are small but significant discrepancies in the assessment of the logical structure of the papers.

Table 2 presents the Pearson correlation coefficients for each evaluation criterion, highlighting the relationship between manual and system-generated evaluations. The correlation coefficients for all criteria ranged from 0.80 to 0.88, indicating a strong positive correlation in all cases. This result suggests that the system is generally consistent with manual evaluations. However, the lower correlation observed for the consistency criterion ($r = 0.80$) may point to the need for further adjustments to the algorithm to improve its ability to replicate the evaluation of papers' logic and consistency more accurately.

These results show that while the system is generally effective in replicating teacher evaluations, there are specific areas, particularly in criteria requiring more excellent subjective judgment, where additional adjustments might be necessary to improve precision and reduce variability in the evaluations generated.

B. IMPACT OF THE SYSTEM ON LEARNING OUTCOMES

Two visual methods and detailed statistical analysis were used to assess the system's impact on learning outcomes. A sample of 200 student grades was collected before and

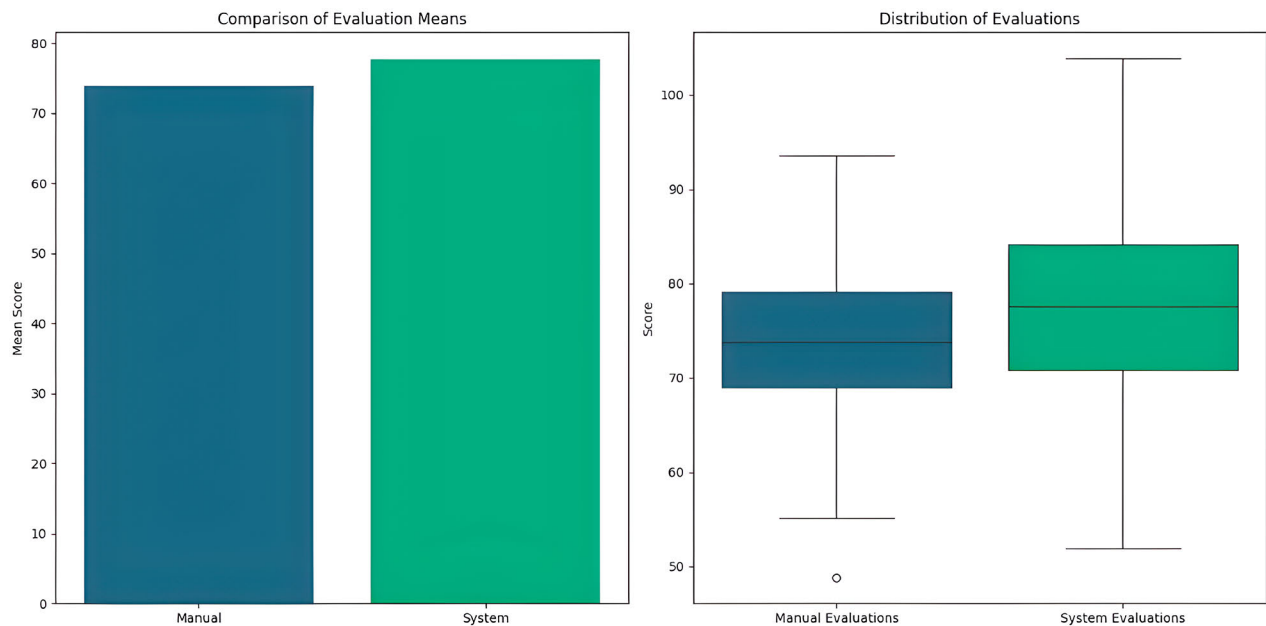


FIGURE 6. Comparison of means and distribution of manual and system-generated evaluations for different criteria.

after the implementation of the system, allowing for a direct comparison of how grades changed following the introduction of this technology. Before any comparative analysis was conducted, the data was standardized to ensure that grades and competency scores were comparable across different assessment sessions and periods. This included normalizing grades where necessary, checking data integrity, and removing any anomalies or missing data that could skew the results.

Statistical analysis focused on quantifying the observed changes in the impact of the system. Descriptive statistics summarizing changes in grades, such as mean, median, and standard deviation, were calculated, providing a clear quantitative context on how student performance varied. To assess the statistical significance of these changes, a student t-test for dependent samples was applied, comparing grades before and after the system. Additionally, the effect size (Cohen's *d*) was calculated, which allowed us to evaluate the magnitude of the system's impact on learning outcomes, concluding that the system produced a moderate but significant improvement in students' academic performance.

The results are presented in Figure 7. The grades were represented in a histogram, which compares the grades' distribution at both times. This histogram shows how the grades shift to the right after the implementation of the system, indicating a general increase in the student's academic performance. In addition, the histogram allows us to visualize the reduction in the dispersion of the grades, suggesting greater consistency in the results. In addition, the system's positive impact on the distribution of grades is identified. The histogram reveals a notable increase in the mean of the grades after the implementation of the

TABLE 3. Statistical summary of ratings before and after implementation of the system.

Metric	Before System	After System
Mean	70.4	75.2
Median	71.0	76.0
Standard Deviation	10.2	8.5
Minimum	45.0	52.0
Maximum	93.0	97.0

system, accompanied by a reduction in the dispersion of the scores, suggesting that the system has helped to homogenize academic performance. This homogenization is particularly important in an educational setting, as it indicates that more students are reaching a consistent level of performance, reducing the gap between students with different initial ability levels.

In parallel, student progress was monitored in two key competencies, selected for their relevance in the assessed educational setting. Progress in critical competencies was measured using an approach based on continuous assessment and quantitative analysis of scores obtained by students in successive assessments over ten sessions. The results were plotted in a line graph.

First, two core competencies were identified as critical to students' academic development in the specific course. These competencies were selected based on their relevance to the course learning objectives and ability to be assessed objectively and quantitatively. The preferred competencies could have included "clarity in written communication" and "coherence in argumentation," for example. Consistent assessments were designed and administered in each of the ten sessions to measure progress in these competencies.

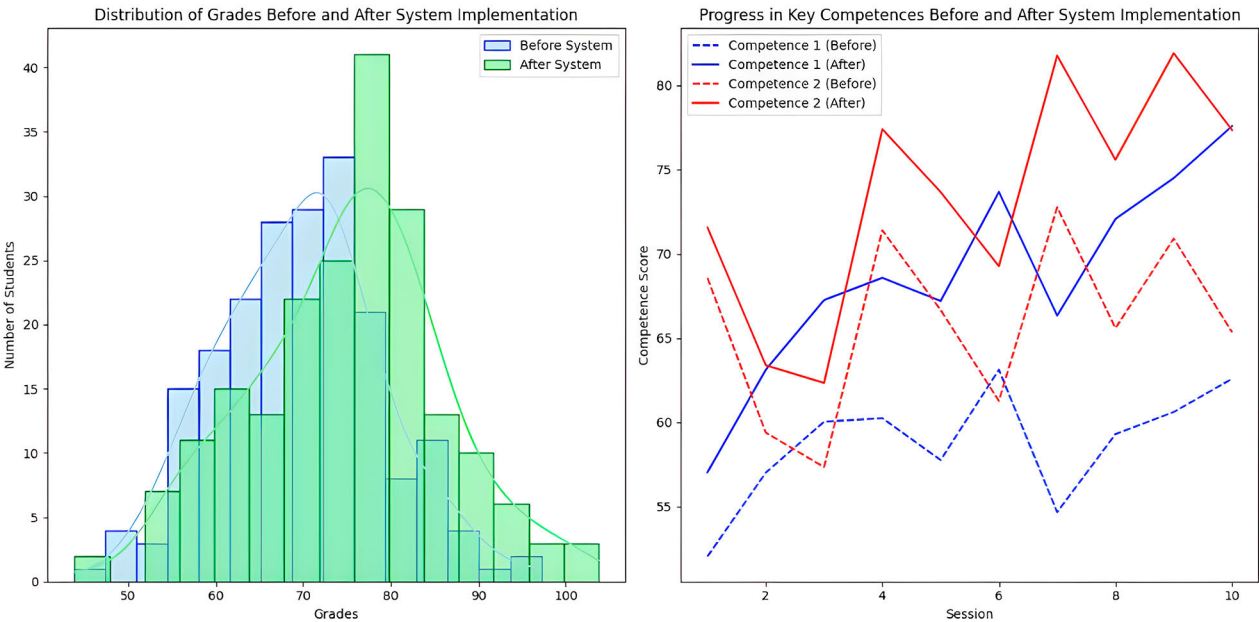


FIGURE 7. Distribution of grades and progress in key competencies before and after the implementation of the system.

These assessments were designed to be comparable, so the scores obtained in each session accurately reflected the development of the corresponding competency over time. The assessment criteria were kept constant across sessions to ensure consistency in measuring progress.

In each session, the scores obtained by students in the assessments of the critical competencies were collected. These scores were recorded and organized in a time series, allowing detailed monitoring of each student’s progress in the selected competencies. Scores were standardized when necessary to ensure comparability between different sessions.

Table 3 summarizes the descriptive statistics for scores. The mean score increased from 70.4 to 75.2, and the median moved from 71.0 to 76.0, indicating that the improvement in scores was widespread and not limited to a few students. The reduction in the standard deviation, from 10.2 to 8.5, suggests a decrease in performance variability, indicating that the system helped level out differences among students of different abilities.

Table 4 presents the results of the student t-test and confirms the significance of the observed improvements. The mean difference of 4.8 points in grades before and after the system is statistically significant, with a p-value less than 0.001, indicating that the probability of this improvement being due to chance is extremely low. Furthermore, the effect size of 0.51 means that the magnitude of this improvement is moderate, suggesting that the system has had a tangible impact on improving academic performance.

C. SYSTEM SATISFACTION AND USABILITY

The satisfaction and usability analysis of the system was carried out by implementing a structured questionnaire designed to evaluate different aspects of the system, such

TABLE 4. Student t-test results and effect size for different competence comparisons.

Comparison	Mean Difference	t-Value	Degrees of Freedom (df)	p-Value	Effect Size (Cohen's d)
Before vs. After (Overall Grades)	4.8	5.12	199	<0.001	0.51
Before vs. After (Clarity Competence)	3.2	4.75	199	<0.001	0.43
Before vs. After (Coherence Competence)	2.7	3.90	199	<0.001	0.35
Before vs. After (Originality Competence)	4.1	5.40	199	<0.001	0.50

as ease of use, efficiency in performing tasks, clarity of the interface, and overall satisfaction. The questionnaire used a five-point Likert scale, where one represented “Very dissatisfied” and 5 “Very satisfied.” Two hundred responses were collected and equally distributed between students and teachers. In addition, semi-structured interviews were conducted to obtain qualitative insights into the system’s user experience. Subsequently, a Pearson correlation analysis was performed to determine the relationship between the usability variables and overall satisfaction.

The survey results, represented in Figure 8, show that most users found the system satisfactory in ease of use, efficiency in performing tasks, and clarity of the interface. In particular, 45% of respondents rated ease of use as “Satisfied,” while 20% rated it as “Very Satisfied.” These results indicate that the system meets most users’ expectations regarding its operability. However, a small percentage of

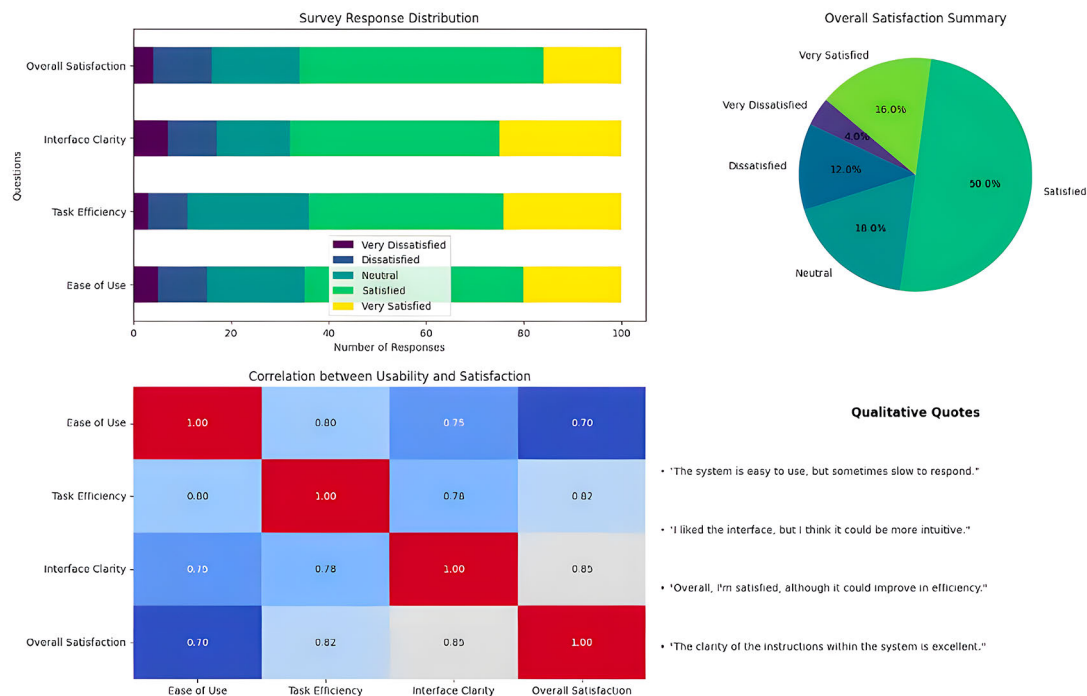


FIGURE 8. Distribution of system satisfaction and usability responses.

TABLE 5. Student t-test results and effect size for different competence comparisons.

	Ease of Use	Task Efficiency	Interface Clarity	Overall Satisfaction
Ease of Use	1.00	0.80	0.75	0.70
Task Efficiency	0.80	1.00	0.78	0.82
Interface Clarity	0.75	0.78	1.00	0.85
Overall Satisfaction	0.70	0.82	0.85	1.00

users, represented by the 5% who rated ease of use as “Very Dissatisfied,” suggests that specific areas of improvement could be addressed in future system iterations.

Overall satisfaction with the system, as presented in the pie chart, indicates that 66% of users reported being satisfied or very satisfied with the system, while 16% expressed dissatisfaction. This chart highlights the widespread acceptance of the system among users. However, it suggests the need to focus on elements causing dissatisfaction, as explored in the qualitative interviews.

The Pearson correlation analysis, the results of which are found in Table 5, reveals a moderate positive correlation between ease of use and overall satisfaction ($r = 0.70$, $p < 0.05$), as well as between task completion efficiency and overall satisfaction ($r = 0.82$, $p < 0.05$). Furthermore, interface clarity correlated with overall satisfaction ($r = 0.85$, $p < 0.05$). These findings suggest improving interface clarity and system operational efficiency could significantly impact users’ satisfaction.

In the qualitative quotes obtained, one user commented, “The system is easy to use, but is sometimes slow to respond,” which is consistent with the survey findings in terms of satisfaction with ease of use but also points to a concern about the speed of the system. Another user noted, “I liked the interface, but I think it could be more intuitive,” reinforcing the positive correlation between interface clarity and overall satisfaction. These qualitative perceptions provide insight into areas where the user experience could be improved.

D. EFFECTIVENESS OF AUTOMATIC SYLLABLE GENERATION

The effectiveness of automatic syllabus generation was analyzed by directly comparing syllabi generated automatically by the system and syllabi designed manually by subject matter experts. The process began with developing an automatic syllabus for a specific course, using previously defined parameters, such as the subject matter, learning objectives, and number of sessions. This syllabus was then compared to an equivalent syllabus designed manually for the same subject matter and under the same learning objectives.

Different sections of the syllabus, including the general organization of the sessions, the proposed activities, and the suggested materials and resources, were selected to ensure an efficient evaluation. These elements were evaluated in terms of their structure, detail, clarity, and relevance to the course objectives. Comparative tables were used to organize

TABLE 6. Comparison of the general structure of the automatically and manually generated syllable.

Aspect	Automatically Generated Syllabus	Manually Designed Syllabus
Unit/Topic	Unit 1: Introduction to Data Science	Unit 1: Introduction to Data Science
Session Organization	Three sessions per topic (pre, during, post)	Two sessions per topic (pre- and during)
Details of Activities	Specific activities for each session	General activities, not divided by session
Estimated Time	Specified for each activity	Total time per topic, not broken down
Materials and Resources	Detailed listings per activity	Mentioned in general

and visualize the differences and similarities between both approaches.

In parallel, qualitative evaluations were collected from subject matter experts, who analyzed the automatically generated syllabi compared to the manual ones, providing feedback on the quality and consistency of the content. Additionally, surveys were conducted with instructors who used the automatically generated syllabi in their courses to collect data on their perception of the usefulness, clarity, and relevance of these syllabi in educational practice.

Table 6 presents a detailed comparison of the overall structure of the automatically generated and manually designed syllabi. The analysis revealed that, in terms of session organization, the automatically generated syllabus tends to offer a greater segmentation of activities into pre-session, session, and post-session, which is not always observed in manual syllabi. This more granular approach allows for more detailed planning of educational activities, ensuring that each instruction phase is delineated. Furthermore, it was observed that the automatically generated syllabus provides more specific time estimates for each activity, which is beneficial for time management by instructors and students. However, in some cases, manual syllabi showed greater flexibility in session organization, which can be advantageous in educational environments that require constant adaptations. This difference suggests that while autofocus offers precision, it could benefit from greater flexibility to adjust to changing classroom needs.

Table 7 compares specific activities and resources between both types of syllabi. It was observed that the activities proposed in the automatically generated syllabus tend to be more detailed, with clear instructions on the steps to follow and the resources needed. For example, in the pre-reading activity on Data Science, the automatic syllabus specifies a specific article, the estimated time for its reading, and the resources to be used. In contrast, the manual syllabus suggests a general reading without detailing the resources or the time needed. This level of detail in the automatic syllabus is beneficial for teachers looking for a complete and structured guide. Still, it can limit the creativity or flexibility that some prefer in the manual design. However, in more open activities, such as essays or group discussions, the manual

TABLE 7. Detailed comparison of activities and resources between automatically and manually generated syllabus.

Session	Proposed Activity	Automatically Generated Syllabus	Manually Designed Syllabus
Pre-Session	Pre-reading on Data Science	Reading a provided article	Recommended reading from a textbook
Session 1	Group discussion on the reading	Group discussion, 30 minutes	Group discussion, no time specified
Post-Session	Short essay on the importance of data	Essay, 60 minutes, computer use	Essay, no time or resources specified
Pre-Session 2	Video on the history of Data Science	Watch the video and take notes, 30 minutes	View the suggested video; there are no clear instructions

TABLE 8. Qualitative evaluation of the clarity and relevance of activities in automatically and manually generated syllabus.

Activity	Clarity (Auto)	Clarity (Manual)	Relevance (Automatic)	Relevance (Manual)
Pre-Session Reading	High	Moderate	High	High
Session 1 Discussion	High	Moderate	High	High
Post-Session Essay	High	Low	Moderate	High
Pre-Session 2 Video	High	Moderate	High	Moderate

syllabus allows for more excellent adaptation to the specific needs of the student group.

Table 8 provides a qualitative assessment of the clarity and relevance of the activities proposed in both syllabi. The results show that the automatically generated activities are generally more transparent in their description, facilitating understanding for teachers and students. This was reflected in the high clarity rating in the automatic syllabus for activities such as group discussion and post-session essays. However, the relevance of the activities varied depending on the specific activity. While clearly defined and structured activities, such as readings and discussions, showed high significance in both syllabi, more open-ended activities, such as essays, were rated as more relevant in the manual syllabus, where teachers can adjust topics and approaches according to the classroom context. This suggests that, although the automatically generated system is highly effective in creating structured and detailed activities, it could benefit from options that allow teachers to customize activities that require a more creative or adaptive approach.

E. ADJUSTMENTS AND IMPROVEMENTS IMPLEMENTED

The evaluation process of the adjustments implemented in the system was carried out in several phases, focusing on measuring the precision of the assessments, the relevance of the generated syllabi, and the system’s overall usability. To obtain reliable data, measurements were made in four different periods: before the adjustments, one month after, three months after, and six months after implementation.

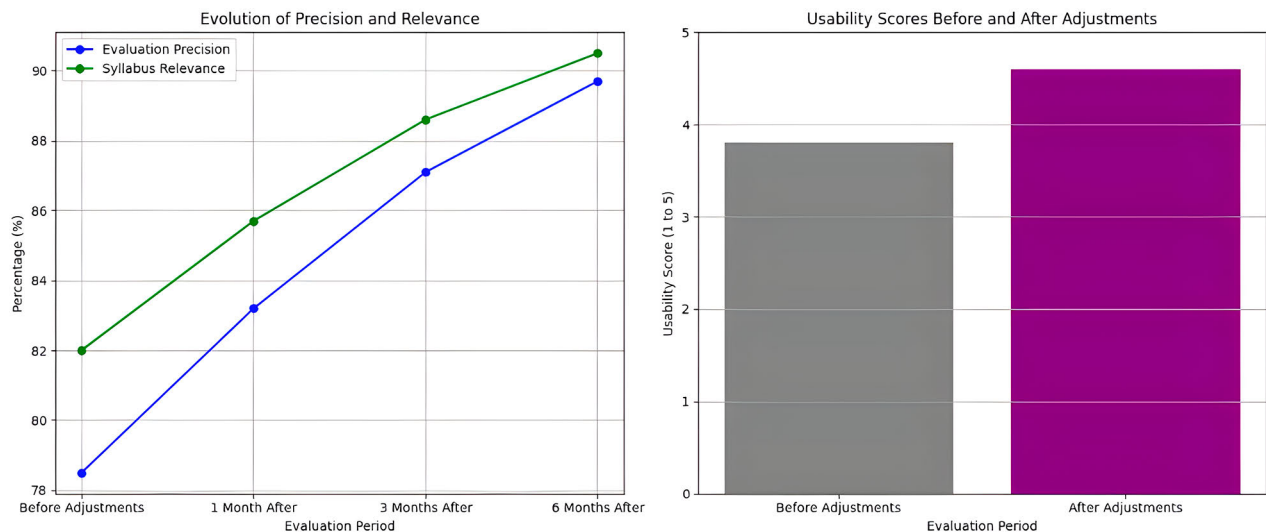


FIGURE 9. Evolution of assessment precision, syllable relevance, and system usability after adjustments.

These benchmarks allowed us to assess the immediate impact of the adjustments and the sustainability of the improvements over time.

Data was collected by recording the system's assessments and comparing them with manual evaluations by experts. Additionally, usability surveys were conducted among users, including teachers and students, to capture their perceptions of the system's ease of use and effectiveness after the adjustments. These surveys used a five-point Likert scale to quantify usability, allowing a clear comparison between the different evaluation periods.

The system adjustments significantly impacted key metrics, as evidenced in Figure 9. The line graph shows a positive evolution in the precision and relevance of the generated assessment syllabi. Before the adjustments, the precision of the assessments was 78.5%, increasing steadily to reach 89.7% six months after the adjustments. This 14.3% improvement suggests that the changes in the assessment algorithms, particularly those that optimized the contextual interpretation of the rubrics, were highly effective.

Similarly, the relevance of the syllabi experienced a notable increase, going from 82.0% before the adjustments to 90.5% six months later. This positive trend indicates that the customization options introduced in the system allowed teachers to better adapt the content to their specific courses, resulting in a more favorable perception of the automatically generated syllabi. The system's ability to maintain this improvement over time indicates the robustness of the adjustments implemented.

The bar chart compares the system's usability before and after the adjustments, showing an increase in the average score from 3.8 to 4.6. This increase in usability is particularly relevant, as it reflects the improvement in the user experience, a critical factor for the system's adoption and long-term success. The main contributors to this improvement were the more intuitive interface and improvements in operational

TABLE 9. Qualitative evaluation of the clarity and relevance of activities in automatically and manually generated syllabus.

Metric	Before Adjustments	After Adjustments	% Change
Assessment Precision (%)	78.5	89.7	+14.3%
Syllabus Relevance (%)	82.0	90.5	+10.4%
Usability (Average Score)	3.8	4.6	+21.1%

efficiency, such as the reduction in loading and processing times.

Table 9 summarizes key metrics before and after the adjustments, highlighting the most significant improvements. The precision of assessments and relevance of syllabi showed the most substantial changes, confirming the effectiveness of the adjustments implemented. Usability, which improved by 21.1%, also stands out as an area of success, underlining the importance of a well-designed user interface and a smooth experience.

V. DISCUSSION

The analysis of our study's results reflects significant advances in the automation of educational processes, particularly in the generation of syllabi and the automatic evaluation of academic works. These advances align with previous findings from studies by Rathi et al. [12] and Liu [13], who demonstrated that precision, personalization, and usability are critical for the acceptance and effectiveness of automated systems in education. Implementing iterative adjustments in our system, based on user feedback, has proven to be a practical approach to improve these key metrics, aligning with the trends observed in recent literature [14].

Implementing the elements in our system was carried out meticulously to improve the evaluations' precision, the

generated syllabi's relevance and the system's usability. Adjustments were made to the text processing algorithms, allowing the system to interpret the context of the rubrics provided by teachers more effectively. This significantly improved assessment precision, which increased by 14.3% after the adjustments, as evidenced in Table 9. This improvement is crucial because assessment precision is critical to maintaining student confidence and ensuring grades accurately reflect their skills and knowledge.

Furthermore, the relevance of the generated syllabi also showed a notable increase, going from 82.0% before the adjustments to 90.5% six months after their implementation. This result suggests that the changes introduced to better customize the generated content were effective. The ability of instructors to adapt syllabi to the specific needs of their courses is essential to ensure that the content is relevant and aligned with the learning objectives. This personalization not only improves the relevance of the content but also contributes to higher user satisfaction, as reflected by the increase in the average usability score, which rose from 3.8 to 4.6, as shown in Figure 9.

Our work contributes to innovation in automated education by demonstrating that an iterative approach based on continuous user feedback can lead to sustainable improvements in educational systems' precision, relevance, and usability. This methodology is applicable in the context of automatic syllabus and assessment generation. Still, it can also be extended to other domains where automation plays a crucial role in improving the efficiency and effectiveness of the educational process [35]. Integrating these adjustments continuously ensures that the system evolves and adapts to the changing needs of users, which is essential for its long-term adoption. However, it is essential to acknowledge the potential limitations of the results obtained. One of the main limitations of our study is the system's dependence on the quality and consistency of the rubrics provided by teachers. While the system has proven effective in interpreting well-structured rubrics, its performance can be compromised when rubrics are ambiguous or lack a clear focus [36]. This could lead to variations in the precision of assessments and, therefore, affect the reliability of the results.

Another significant limitation is the assumption that all users have a homogeneous level of familiarity with the technology used. Although efforts were made to improve the system's usability, it is possible that some users, especially those with less experience in technology, still find it challenging to interact with the system efficiently. This could limit the applicability of the results in educational settings where there is an excellent diversity in users' technological skills.

Furthermore, the study was conducted in a controlled environment, which implies that the conditions under which the adjustments were evaluated might not entirely reflect the variability and complexity of natural educational environments. Restrictions imposed by this controlled environment, such as selecting a specific set of courses and limiting the

variety of rubrics used, might have influenced the results obtained. Consequently, the generalization of these findings to other educational contexts should be done with caution.

Finally, the interpretation of the results is also affected by the predominant quantitative approach in our analysis. While the precision, relevance, and usability metrics provide an objective view of the impact of the implemented adjustments, they may not fully capture users' subjective experiences and perceptions. Future studies could benefit from further integrating qualitative methodologies to explore how users experience and perceive the system, which could offer new insights into improving its effectiveness and acceptance.

VI. CONCLUSION

This study has explored the potential of automation in education, specifically through automatic syllabus generation and automated assessment of academic papers. By implementing an iterative and user-centered approach, we addressed some of the main limitations identified in previous research, such as lack of personalization and imprecision in complex assessments. The results show quantifiable improvements in key metrics and highlight the importance of integrating continuous user feedback to optimize the system's functionality and usability.

The system showed significant improvements in the precision of the assessments and the relevance of the generated syllabi. This demonstrates that the adaptations made to the processing algorithms and content personalization are viable and effective in practice. Furthermore, the system's usability improvement, reflected in higher user ratings, underlines the importance of an intuitive interface and efficient performance for the acceptance and sustained use of technology in educational settings.

A key finding of this study is that when properly tuned and adapted, automated technology can significantly improve the educational process's efficiency and quality. This is especially relevant in contexts where the administrative burden on teachers can be reduced through automation, allowing them to focus more on teaching and direct interaction with students. The system's ability to maintain and improve key metrics over time also suggests that an adaptive approach is essential for the sustainability and scalability of these technological solutions in education.

However, it is essential to recognize that our study also faced limitations. The system's reliance on the quality and clarity of rubrics teachers provide underscores the need for a standardized assessment framework. Variability in the quality of these rubrics could lead to inconsistencies in results, raising the need to develop additional mechanisms that allow the system to handle these variations better. Furthermore, diversity in technological familiarity among users is a challenge that must be addressed through targeted training to maximize the system's impact.

Another aspect that deserves attention is the inherent limitation of the controlled context in which the study was

conducted. While the results are promising, the applicability of these findings to more diverse and less controlled educational settings requires further evaluation. This is especially relevant when technological resources are limited, or there is excellent variability in users' digital competencies. The scalability of the system and its adaptation to different cultural and pedagogical contexts are areas that need to be further explored in future research.

Looking ahead, several opportunities to expand and improve the system present themselves. One promising direction is the integration of qualitative methodologies to capture users' subjective perceptions more deeply, which could offer new insights into optimizing human-technology interaction in the educational domain. Furthermore, expanding the system to include automatic adaptation capabilities based on student performance could increase its effectiveness, allowing for even more precise personalization of academic content.

REFERENCES

- [1] M. Revilla-León, M. Gómez-Polo, S. Vyas, A. B. Barmak, G. O. Gallucci, W. Att, M. Özcan, and V. R. Krishnamurthy, "Artificial intelligence models for tooth-supported fixed and removable prosthodontics: A systematic review," *J. Prosthetic Dentistry*, vol. 129, no. 2, pp. 276–292, Feb. 2023, doi: [10.1016/j.prosdent.2021.06.001](#).
- [2] J. Zang, J. Gowthami, and C. Anilkumar, "Adaptive artificial intelligent technique to improve acquisition of knowledge in the educational environment," *J. Interconnection Netw.*, vol. 22, Apr. 2022, Art. no. 2143013, doi: [10.1142/s0219265921430131](#).
- [3] K. Luetsch, G. Wong, and D. Rowett, "A realist synthesis of educational outreach visiting and integrated academic detailing to influence prescribing in ambulatory care: Why relationships and dialogue matter," *BMJ Quality Saf.*, vol. 33, no. 1, pp. 43–54, Jan. 2024, doi: [10.1136/bmjqs-2022-015498](#).
- [4] M. Andronie, M. Iatagan, C. Uță, I. Hurloiu, A. Dijmărescu, and I. Dijmărescu, "Big data management algorithms in artificial Internet of Things-based fintech," *Oeconomia Copernicana*, vol. 14, no. 3, pp. 769–793, Sep. 2023, doi: [10.24136/oc.2023.023](#).
- [5] W. Qiao and C. Yijun, "The role of live transcripts in synchronous online L2 classrooms: Learning outcomes and learner perceptions," *Educ. Inf. Technol.*, vol. 28, no. 11, pp. 14783–14804, Nov. 2023, doi: [10.1007/s10639-023-11784-8](#).
- [6] O. Kolade, A. Owoseni, and A. Egbetokun, "Is AI changing learning and assessment as we know it? Evidence from a ChatGPT experiment and a conceptual framework," *Heliyon*, vol. 10, no. 4, Feb. 2024, Art. no. e25953, doi: [10.1016/j.heliyon.2024.e25953](#).
- [7] N. Wlömert, D. Papies, M. Clement, and M. Spann, "Rejoinder on 'frontiers: The interplay of user-generated content, content industry revenues, and platform regulation: Quasi-experimental evidence from YouTube,'" *Marketing Sci.*, vol. 43, no. 1, pp. 16–19, 2024, doi: [10.1287/mksc.2023.0369](#).
- [8] M. Alier, F.-J. García-Peñalvo, and J. D. Camba, "Generative artificial intelligence in education: From deceptive to disruptive," *Int. J. Interact. Multimedia Artif. Intell.*, vol. 8, no. 5, p. 5, 2024, doi: [10.9781/iji-mai.2024.02.011](#).
- [9] R. L. Versalle, B. R. Todd, N. W. Chen, and D. E. Turner-Lawrence, "Early emergency medicine milestone assessment for predicting first-year resident performance," *MedEdPORTAL*, vol. 20, p. 11386, Feb. 2024, doi: [10.15766/mep_2374-8265.11386](#).
- [10] D. Sengupta, S. S. Khan, S. Das, and D. De, "FedEL: Federated education learning for generating correlations between course outcomes and program outcomes for Internet of Education Things," *Internet Things*, vol. 25, Apr. 2024, Art. no. 101056, doi: [10.1016/j.iot.2023.101056](#).
- [11] R. Ragasudha and M. Saravanan, "Secure automatic question paper generation with the subjective answer evaluation system," in *Proc. Int. Conf. Smart Technol. Syst. Next Gener. Comput. (ICSTSN)*, Mar. 2022, pp. 1–5, doi: [10.1109/ICSTSN53084.2022.9761323](#).
- [12] S. Rathi, P. Wawage, and A. Kulkarni, "Automatic question generation from textual data using NLP techniques," in *Proc. Int. Conf. Emerg. Smart Comput. Informat. (ESCI)*, Mar. 2023, pp. 1–7, doi: [10.1109/ESCI56872.2023.10100278](#).
- [13] C. Liu, "Automating student assessment using digital data to improve education management effectiveness in higher education institutions," *Educ. Inf. Technol.*, vol. 29, no. 2, pp. 1885–1901, Feb. 2024, doi: [10.1007/s10639-023-11898-z](#).
- [14] E. Sing Kiang Siew, Z. Yan Chong, S. Nah Sze, and R. Hardi, "Streamlining attendance management in education: A web-based system combining facial recognition and QR code technology," *J. Adv. Res. Appl. Sci. Eng. Technol.*, vol. 33, no. 2, pp. 198–208, Nov. 2023, doi: [10.37934/araset.33.2.198208](#).
- [15] J. Wilson, F. Zhang, C. Palermo, T. C. Cordero, M. C. Myers, H. Eacker, A. Potter, and J. Coles, "Predictors of middle school students' perceptions of automated writing evaluation," *Comput. Educ.*, vol. 211, Apr. 2024, Art. no. 104985, doi: [10.1016/j.compedu.2023.104985](#).
- [16] V. Alto, *Modern Generative AI with ChatGPT and OpenAI Models: Leverage the capabilities of OpenAI's LLM for productivity and innovation with GPT-3 and GPT-4*. Packt Publishing, Jun. 2023, pp. 15–55.
- [17] A. K. Dubey, "AI LLM integration," *Int. J. Res. Appl. Sci. Eng. Technol.*, vol. 11, no. 12, pp. 1842–1849, Dec. 2023, doi: [10.22214/ijraset.2023.57731](#).
- [18] N. George, *Mastering Django: Core*. Birmingham, U.K.: Packt, 2016.
- [19] Amazon Web Services—AWS, *Data Lakes and Analytics on AWS—Amazon Web Services*, AWS, Seattle, WA, USA, Sep. 2024.
- [20] F. Foping, J. Walsh, and D. Roche, "Design and implementation of a private RESTful API to leverage the power of an eCommerce platform," in *Proc. Int. Conf. Inf. Integr. Web-Based Appl. Services*, Dec. 2013, pp. 681–685, doi: [10.1145/2539150.2539251](#).
- [21] K. Wilkosinska, A. Aderhold, H. Graf, and Y. Jung, "Towards a common implementation framework for online virtual museums," in *Design, User Experience, and Usability. Health, Learning, Playing, Cultural, and Cross-Cultural User Experience* (Lecture Notes in Computer Science). Berlin, Germany: Springer-Verlag, 2013, doi: [10.1007/978-3-642-39241-2_36](#).
- [22] O. Raitio, A. Juslén, P. Sirkkiä, A. Lehtikoinen, M. Tähtinen, E. Piirainen, and V.-M. Riihikoski, "Notebook: Customizable web forms for recording observations," *Biodiversity Inf. Sci. Standards*, vol. 3, pp. 1–3, Aug. 2019, doi: [10.3897/biss.3.39150](#).
- [23] J. A. Greene, L. A. Hutchison, L. J. Costa, and H. Crompton, "Investigating how college students' task definitions and plans relate to self-regulated learning processing and understanding of a complex science topic," *Contemp. Educ. Psychol.*, vol. 37, no. 4, pp. 307–320, 2012, doi: [10.1016/j.cedpsych.2012.02.002](#).
- [24] T. Hao, Z. Huang, L. Liang, H. Weng, and B. Tang, "Health natural language processing: Methodology development and applications," *JMIR Med. Informat.*, vol. 9, no. 10, Oct. 2021, Art. no. e23898, doi: [10.2196/23898](#).
- [25] R. Gruetzmacher, "The power of natural language processing," *Harvard Bus. Rev.*, vol. 100, no. 2, pp. 124–133, Apr. 2022.
- [26] A. S. Alammery, "BERT models for Arabic text classification: A systematic review," *Appl. Sci.*, vol. 12, no. 11, p. 5720, Jun. 2022, doi: [10.3390/app12115720](#).
- [27] L. Floridi and M. Chiriatti, "GPT-3: Its nature, scope, limits, and consequences," *Minds Mach.*, vol. 30, no. 4, pp. 681–694, Dec. 2020, doi: [10.1007/s11023-020-09548-1](#).
- [28] Z. Xu, W. Zou, and D. Lin, "Generation method of hierarchical pronunciation font for hearing-impaired children based on artificial intelligence," in *Proc. IEEE Int. Conf. Adv. Electr. Eng. Comput. Applications (AEECA)*, Aug. 2020, pp. 591–595, doi: [10.1109/AEECA49918.2020.9213534](#).
- [29] Z. Sun, Z. Zhang, and M. Zhang, "Bilstm personalised-style poetry generation algorithm," in *Proc. IEEE 3rd Int. Conf. Power, Electron. Comput. Appl. (ICPECA)*, Jan. 2023, pp. 836–840, doi: [10.1109/ICPECA56706.2023.10075878](#).
- [30] L. D. M. Ortiz-Aguilar, M. Carpio, J. A. Soria-Alcaraz, H. Puga, C. Díaz, C. Lino, and V. Tapia, "Training OFF-line hyperheuristics for course timetabling using K-folds cross validation," *Math. Program. Softw.*, vol. 8, no. 3, pp. 1–8, Oct. 2016.
- [31] "OpenAI's ChatGPT and the prospect of limitless information: A conversation with ChatGPT," *J. Int. Affairs*, vol. 75, no. 1, 2022.

- [32] L. Rogers, S. Hughes Spence, P. Aivalli, A. De Brún, and E. McAuliffe, "A systematic review critically appraising quantitative survey measures assessing power dynamics among multidisciplinary teams in acute care settings," *J. Interprofessional Care*, vol. 38, no. 1, pp. 156–171, Jan. 2024, doi: [10.1080/13561820.2023.2168632](https://doi.org/10.1080/13561820.2023.2168632).
- [33] R. Roberts, "Qualitative interview questions: Guidance for novice researchers," *Qualitative Rep.*, vol. 25, no. 9, pp. 3185–3203, Sep. 2020, doi: [10.46743/2160-3715/2020.4640](https://doi.org/10.46743/2160-3715/2020.4640).
- [34] E. M. Monje, E. V. Cano, and M. Fernández, "Peer assessment of language learning resources in virtual learning environments with e-rubrics," *Int. J. Technol. Enhanced Learn.*, vol. 6, no. 4, p. 321, 2014, doi: [10.1504/ijtel.2014.069018](https://doi.org/10.1504/ijtel.2014.069018).
- [35] B. Das, M. Majumder, A. A. Sekh, and S. Phadikar, "Automatic question generation and answer assessment for subjective examination," *Cognit. Syst. Res.*, vol. 72, pp. 14–22, Mar. 2022, doi: [10.1016/j.cogsys.2021.11.002](https://doi.org/10.1016/j.cogsys.2021.11.002).
- [36] M. Masrul, B. H. Wicaksono, S. Yuliani, S. Erliana, and U. Rasyidah, "The dynamic influence of interactive feedback on elevating EFL students' writing skills," *Stud. English Lang. Educ.*, vol. 11, no. 1, pp. 133–152, Jan. 2024, doi: [10.24815/siele.v11i1.30836](https://doi.org/10.24815/siele.v11i1.30836).



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